DRILL 15
VENTILATION

NFPA 1001 Level 1 (JPR 4.3.11, 4.3.12) / WorkSafeNB Reg 91-191

POSITIVE PRESSURE VENTILATION (PPV) & HYDRAULIC VENT

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Rescue | 🦺 **PPE:** Full Structural Turnout Gear, Eye & Hearing Protection

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Ventilation Crew

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must understand how to position and operate a **Positive Pressure Ventilation (PPV)** fan (gas-powered or electric battery) to establish a "cone of air" covering the entry doorway, coordinate with an exhaust opening to rapidly clear smoke and toxic heat, and execute **Hydraulic Ventilation** using a wide fog stream from an interior attack line.

📦 TOOLS & APPARATUS SETUP

- Apparatus PPV fan (gasoline or electric PPV).
- Attack hose line (1.75" / 45mm) with combination fog nozzle.
- Station bay doors, open hallway, or training structure.
- Hearing protection and eye protection.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Carbon Monoxide (CO) Hazard:** Gasoline-powered fans generate high concentrations of CO in their exhaust. Never point a gas PPV fan into a structure without an open exhaust opening, and monitor interior air with a 4-gas detector.

⚠️ **Hearing Protection:** PPV fans generate high decibel noise. Wear earplugs or headsets during operation.

🛑 **THE STOP RULE: Shout "STOP!" immediately if an uncoordinated fan is energized before interior attack crews confirm water is on the fire, or if smoke starts churning backwards out the intake.**

PHASE 1: PPV PRINCIPLES & THE "CONE OF AIR" (15 MINUTES) — SLOW DRY RUN

1. **The 3 Rules of PPV:**
   - **Rule 1: Coordinate with Attack.** Never start a PPV fan during an active interior structural fire until the interior attack team confirms: *"Fire knocked down, ready for ventilation!"* Introducing high-velocity air into an unknocked-down fire causes rapid thermal runaway (blowtorch effect)!
   - **Rule 2: Exhaust Opening First.** An exhaust opening (window or exterior door) in the fire compartment must be opened **BEFORE** the fan is brought to full throttle. The exhaust opening should be approximately the same size or slightly larger than the intake door (1:1 to 1:1.5 ratio).
   - **Rule 3: Seal the Intake ("Cone of Air").** Position the fan 4 to 10 feet back from the doorway. Tilt the fan shroud so the pressurized air stream forms a visible cone covering all four corners of the door frame.
2. **Checking the Seal:** Walk across the doorway with a gloved hand. You should feel pressurized air pushing inward at all four corners, with no smoke or air swirling backward around the edges.

DRILL 15: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Ventilation • Practical Application

PHASE 2: ELECTRIC VS GAS FAN OPERATION (15 MINUTES) — BUDDY COACHING

1. **Gas-Powered Fan:**
   - Check fuel level. Set choke, open fuel valve, pull starter cord smoothly.
   - Allow 30 seconds of warm-up before increasing throttle to full RPM.
2. **Battery / Electric Fan:**
   - Verify battery pack charge status.
   - Position at doorway, press power switch, rotate variable speed dial.
   - Highlight benefits: Zero CO exhaust, instant on/off control, quiet operation allowing fireground radio communication.

PHASE 3: HYDRAULIC VENTILATION (INTERIOR FOG STREAM) (20 MINUTES) — TIMED / STRESS

1. **The Principle:** A flowing fog stream moving through an open window creates a low-pressure Venturi effect that pulls thousands of cubic feet of smoke and steam out behind it.
2. **Execution Steps:**
   - Interior nozzle team positions 2 to 3 feet back from an open exterior window.
   - Adjust fog nozzle to a **wide-fog pattern (60 to 90 degrees)**.
   - Direct the stream so the outer edge of the fog pattern covers 85% to 90% of the window opening.
   - Open the nozzle bail fully.
   - Observe the dramatic drop in interior smoke as air and steam are drafted out the window.
   - Flow for 60 seconds, shut down smoothly, and evaluate visibility.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ✗ *Pressurizing without an exhaust:* Turning on a fan with all windows closed creates extreme pressure, forcing superheated smoke and fire into concealed ceiling chases and attics.
- ✗ *Positioning fan too close to the door:* If placed 1 foot from the door, the air cone does not expand enough to cover the upper door corners, allowing smoke to cycle back into the firefighter's face.
- ✗ *Ignoring wind direction:* Always try to ventilate **with the wind**, not against prevailing head-winds.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did you feel any air escaping around the top corners of the doorway?", "How much quieter was the battery fan compared to the gas unit?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Gas fan fuel topped up; fuel petcock valve shut.
- ☐ Electric fan batteries returned to shore-power chargers.
- ☐ Fan blades and guard wiped clean of dust and soot.
- ☐ Hearing protection stowed.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS